

INPUT price  
total = price \* 1.1  
PRINT total  
// define C=TO-P  
DECLARE temp As float  
DECLARE result As float  
//

## SIMATS ENGINEERING ASSESSMENT TOOL 3

COURSE CODE: CSA0211

COURSE NAME: C Programming

### COURSE OUTCOME COVERED

CO1: Construct structured C programs using constants, variables, data types, and preprocessor directives to address basic computational tasks. (BL2, BL3)

Assessment Tool Weightage: 10%

Total Marks: 100

### QUESTIONS:

Annexure A

MCQ

Code of Conduct:

[192311446]

I Y. Hema Sailahan (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.  
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Y. Hema Sailahan



Q: Given  $\text{vec}(\mathbf{A})$  is the  $\text{vec}$  of an  $m \times n$  matrix and  
 value  $\text{approx}(\mathbf{A})$ , what is the final output?

let  $\mathbf{a} = \text{vec}(\mathbf{A})$   
 print  $\mathbf{a}^T \mathbf{a}$ , ok  
 Output?

```
Q5. Given
value app
int a = 0100;
printf "%x",
a;
Output?
```

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Q5. Geek uses 010 as the value of an integer and prints it using %x. After converting the value appropriately, what will be the final output?

```
int n = 010;  
printf("%x", n);
```

Output?

010 (octal) = 8 (hex)

☒ A) 8

B) A

C) 10

D) Error

Q6. A technician adds a hexadecimal value 0x10 and an octal value 010, then prints the result using %d. What will be the output?

```
printf("%d", 0x10 + 010);
```

Output?

16 + 8 = 24

☒ A) 24

B) 18

C) 20

D) 16

Q7. A variable with value 20 is printed using both %o and %x format specifiers in the same statement. What will be the output?

```
int n = 20;
```

```
printf("%o %x", n, n);
```

Output?

☒ A) 24 14

20 is 24 in octal & 14 in hexadecimal

B) 20 20

C) 24 20

D) Error

Q8. A macro #define A 10+5 is used in an expression A\*2 without parentheses. What will be the output?

```
#define A 10+5
```

```
printf("%d", A*2);
```

Output?

☒ A) 30

10 + 5 \* 2

B) 25

C) 40

D) Error

Q9. A macro #define X 10 is defined, and later a variable with the same name int X = 5; is declared. What will happen when printing X?

```
#define X 10
```

```
int X = 5;
```

```
printf("%d", X);
```

Output?

☒ A) 10

B) 5

C) Error

D) Garbage

A macro name cannot be redeclared as a variable



Q10. A constant variable is declared as "const int x = 10" and then reassigned using "x = x + 5". What will be the result?

```
const int x = 10;
```

```
x = x + 5;
```

Result?

A) 15

B) 10

☒ C) Error

D) Garbage

*A const variable cannot be modified*

Q11. A student divides an integer (5) by a float (2) and prints the result using %f. What will be the output?

```
int a = 5;
```

```
float b = 2;
```

```
printf("%f", a/b);
```

Output?

A) 2.000000

☒ B) 2.500000

C) 2

D) Error

*integer is promoted to float*

Q12. A float variable is initialized as float x = 5/2 and then printed using %f. What will be the output?

```
float x = 5/2;
```

```
printf("%f", x);
```

Output?

A) 2.500000

☒ B) 2.000000

C) 5

D) Error

Q13. A character variable is assigned the value 100 and printed using %c. What will be displayed?

```
char c = 100;
```

```
printf("%c", c);
```

Output?

A) 100

☒ B) d

C) Error

D) Garbage

Q14. A macro, A, is defined and then undefined using #undef. What will happen when attempting to print A?

```
#define A 5
```

```
#undef A
```

```
printf("%d", A);
```

Output?

A) 5

B) 0

*A is undefined after #undef*



Q15. A variable is declared but not initialized, and then printed. What will be the output?  
int a;  
printf("%d", a);

Output?

A) 0

B) Error

☒ C) Garbage value

D) 1

Q16. A character, 'A', is stored in a variable, and 1 is added to it before printing the result as an integer. What will be the output?

char a = 'A';

int b = a + 1;

printf("%d", b);

Output?

☒ A) 66

B) 65

C) A

D) Error

Q17. A macro #define B 2+3 is used in the expression B + 4. What will be the output?

#define B 2+3

printf("%d", B+4);

Output?

☒ A) 9

B) 10

C) 11

D) Error

Q18. A macro #define X 3+2 is used in the expression X\*X without parentheses. What will be the output?

#define X 3+2

printf("%d", X\*X);

Output?

A) 25

B) 11

☒ C) 19

D) Error

Q19. An integer is divided by a float and printed using %f. What will be the output of 10/3?

int a = 10;

float b = 3;

printf("%f", a/b);

What will be printed?

☒ A) 3.333333

B) 3.000000



```

// TAX = 0.05
DECLARE price As float
DECLARE total As float
INPUT price
total = price + (price * TAX)
PRINT total

```

- C) 3  
D) Error

**Q20.** A character 'A' is printed using %d. What will be the output?

```

char ch = 'A';
printf("%d", ch);

```

Output?

- A) A  
B) 65  
C) Error  
D) Garbage

### RUBRICS FOR CALCULATION:

| Criteria                 | Weightage (%) | Level 4 (Exemplary)                                                   | Level 3 (Proficient)                         | Level 2 (Developing)                               | Level 1 (Beginning)                                |
|--------------------------|---------------|-----------------------------------------------------------------------|----------------------------------------------|----------------------------------------------------|----------------------------------------------------|
| Conceptual Understanding | 30%           | Demonstrates thorough understanding; answers all questions accurately | Shows good understanding with few errors     | Partial understanding; several incorrect responses | Lacks understanding; majority of answers incorrect |
| Accuracy of Answers      | 25%           | All answers are correct                                               | Most answers are correct with minor mistakes | Some correct answers; frequent errors              | Mostly incorrect answers                           |
| Application of Knowledge | 20%           | Correctly applies concepts to problem-based questions                 | Applies concepts with minor errors           | Limited ability to apply concepts                  | Unable to apply concepts                           |
| Time Management          | 15%           | Completes quiz well within time efficiently                           | Completes on time with minor delays          | Requires extra time or rushes through questions    | Unable to complete within given time               |
| Question Interpretation  | 10%           | Interprets all questions correctly and responds appropriately         | Misinterprets a few questions                | Often misinterprets questions                      | Unable to understand questions                     |